* XgBoost ML Algorithm

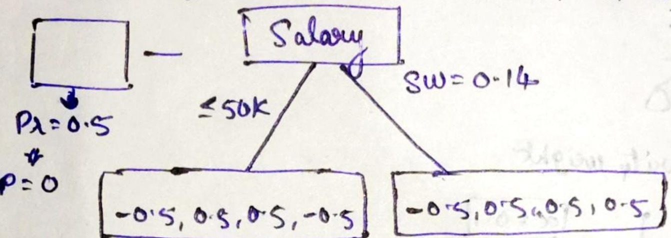
(Classification)

Dataset

$\hat{y} = 0.5$

Salary	Credit	Approval (y)	R_1	$\hat{y}$	R_2
<=50K	B	0	-0.5	0.52	-0.48
<=50K	G	1	0.5	0.58	-0.42
<=50K	G	1	0.5	-	-
>50K	B	0	-0.5	-	-
>50K	G	1	0.5	-	-
>50K	N	1	0.5	-	-
<=50K	N	0	-0.5	-	-

[-0.5, 0.5, 0.5, -0.5, +0.5, 0.5, -0.5]



Similarity weight = 0 Similarity weight = 0.33

Steps:

i. Construct a base model

ii. Construct a Decision Tree with root

iii. Calculate similarity weight

$$= \frac{\sum (\text{Residual})^2}{\sum Pr(1-Pr) + \lambda}$$

iv. Calculate

Similarity weight (LC) $= \dfrac{\sum(\text{Residual})^2}{\sum Pr(1-Pr)}$

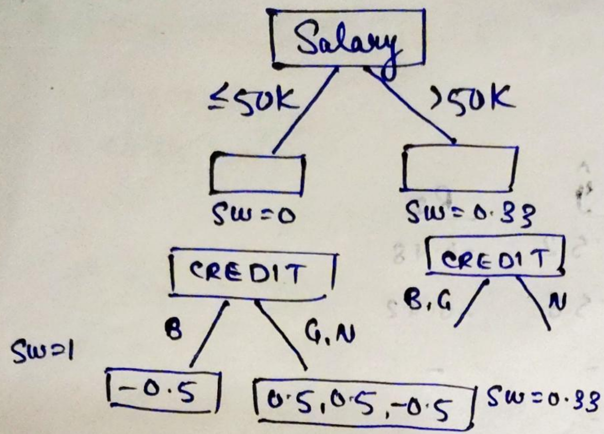
Left child

$$= \frac{[(-\cancel{0.5} + \cancel{0.5} + \cancel{0.5} - \cancel{0.5})^2]}{[0.5(1-0.5) + 0.5(1-0.5) - 0.5(1-0.5) + 0.5(1-0.5)]} = \frac{0}{1} = 0$$

Similarity weight (RC) $= \dfrac{(-\cancel{0.5} + \cancel{0.5} + 0.5)^2}{0.75} = \dfrac{0.25}{0.75} = \dfrac{1}{3} = 0.33$

Gain $= 0 + 0.33 - 0.14 = 0.19$

SW(Root) $= \dfrac{0.25}{1.75} = 0.14$



$$\text{Similarity weight}(L.C) = \frac{0.25}{0.25} = 1$$

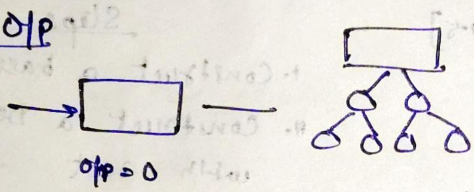
$$SW(RC) = \frac{0.25}{0.75} = 0.33$$

$$\text{Gain} = 1 + 0.33 - 0$$
$$= 1.33$$

Test data

$$\log(\text{odds}) = \log\left(\frac{P}{1-P}\right) = \log\left(\frac{0.5}{0.5}\right) = \log(1) = 0$$

Final O/P



Similarity weight

New Data → $= \sigma(0 + \alpha(1))$ $\boxed{\alpha = 0.1}$

logistic function ⇐ Sigmoid Activation

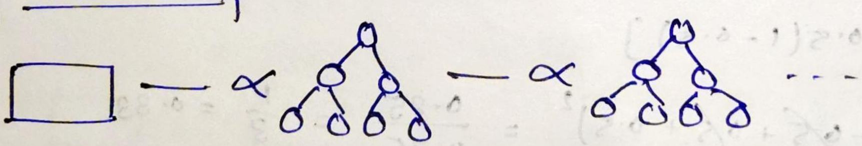
$$= \sigma(0 + (0.1)(1))$$

$$= \frac{1}{1+e^{-0.1}} = 0.52$$

Second Record

$$o/p = \sigma(0 + \alpha(0.33))$$
$$= \sigma(0 + 0.1(0.33))$$
$$= \frac{1}{1+e^{-0.033}} = 0.508$$

Xg Boost Classifier:



$$o/p = \sigma(\text{Base learner} + \alpha_1(DT_1) + \alpha_2(DT_2) + \alpha_3(DT_3))$$